

# ELITE IGCSE ACADEMY

*Student Past Paper Solutions*

## Jun 2025 P1HR Worked Solutions

Jun 2025 | P1HR

33 questions   ■   question image followed by worked solution

PREPARED BY

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STUDENT EDITION

**Worked answers included**

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## PAST PAPER QUESTION

Q#: 1

Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jun 2025 P1HR - Jun 2025 | P1HR

1 Find the lowest common multiple (LCM) of 45 and 70

(Total for Question 1 is 2 marks)

EA

## PAST PAPER SOLUTION

Q#: 1

Marks: 2

TOPIC: **Prime Factors, HCF & LCM**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Jun 2025 P1HR - Jun 2025 | P1HR

**2** The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

..... m

**(1)**

(ii) Write down the upper bound of the length of the ship.

..... m

**(1)**

---

**(Total for Question 2 is 2 marks)**

EA

## PAST PAPER SOLUTION

Q#: 2

Marks: 2

TOPIC: **Rounding, Estimation & Bounds**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 3****Marks: 3**TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

3 Show that  $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

---

(Total for Question 3 is 3 marks)

EA

## PAST PAPER SOLUTION

Q#: 3

Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

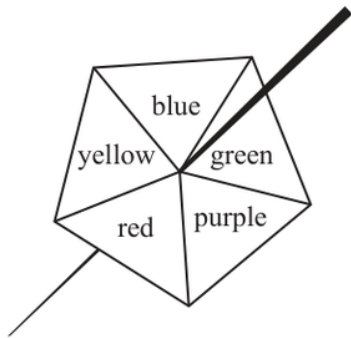
PAST PAPER QUESTION

Q#: 4    Marks: 5

TOPIC: Probability Toolkit

Jun 2025 P1HR - Jun 2025 | P1HR

- 4 Here is a biased 5-sided spinner.  
When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.



The table gives information about the probability of the spinner landing on each colour.

| Colour      | blue | green | purple | red  | yellow |
|-------------|------|-------|--------|------|--------|
| Probability | 0.12 | 0.20  | 0.38   | $4x$ | $x$    |

Sophie spins the spinner once.

- (a) Work out the probability that the spinner lands on blue or on green or on purple.

.....  
(1)

Max spins the spinner 350 times.

- (b) Work out an estimate for the number of times the spinner lands on red.

.....  
(4)

(Total for Question 4 is 5 marks)

## PAST PAPER SOLUTION

Q#: 4

Marks: 5

TOPIC: **Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 5****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

5  $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

(a) List the members of the set

(i)  $A \cup B$

.....

(ii)  $B'$

.....

(2)

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

|               |        |        |             |       |          |
|---------------|--------|--------|-------------|-------|----------|
| $\mathcal{E}$ | $\cap$ | $\cup$ | $\emptyset$ | $\in$ | $\notin$ |
|---------------|--------|--------|-------------|-------|----------|

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i)  $A \cap C = \dots\dots\dots$

(ii)  $13 \dots\dots\dots \mathcal{E}$

(2)

**(Total for Question 5 is 4 marks)**

## PAST PAPER SOLUTION

Q#: 5

Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

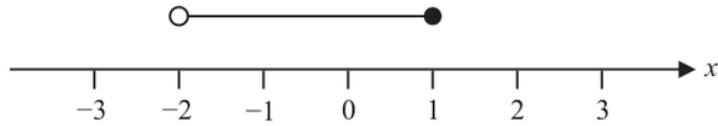
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 6****Marks: 4****TOPIC: Solving Inequalities**

Jun 2025 P1HR - Jun 2025 | P1HR

**6**

- (a) Write down the inequality shown on the number line.

.....  
(2)

- (b) Solve the inequality  $7a - 5 \leq 3a + 28$   
Show clear algebraic working.

.....  
(2)

.....  
(Total for Question 6 is 4 marks)

## PAST PAPER SOLUTION

Q#: 6

Marks: 4

TOPIC: **Solving Inequalities**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 7****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2025 P1HR - Jun 2025 | P1HR

7 Change a speed of  $50x$  metres per second to a speed in kilometres per hour.

..... km/h

---

(Total for Question 7 is 3 marks)

EA

## PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 8****Marks: 5****TOPIC: Factorising**

Jun 2025 P1HR - Jun 2025 | P1HR

**8** (a) Simplify  $a^6 \times a^{10}$ .....  
(1)(b) Simplify  $c^{30} \div c^{12}$ .....  
(1)(c) (i) Factorise  $y^2 - 10y + 21$ .....  
(2)(ii) Hence, solve  $y^2 - 10y + 21 = 0$ .....  
(1)

---

**(Total for Question 8 is 5 marks)**

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## PAST PAPER SOLUTION

Q#: 8

Marks: 5

TOPIC: **Factorising**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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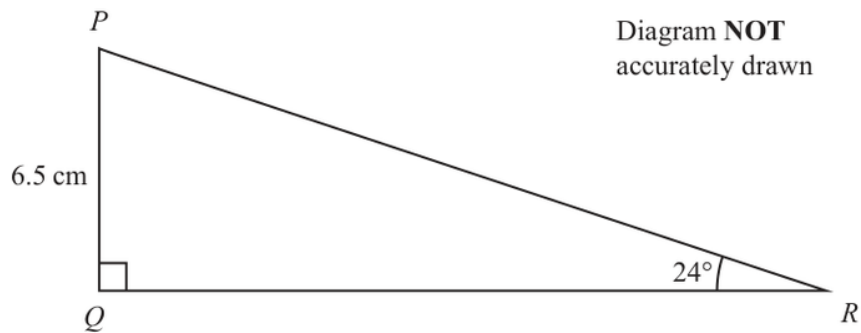
No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 9****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

- 9 The diagram shows triangle  $PQR$



Work out the length of  $QR$   
Give your answer correct to 3 significant figures.

..... cm

---

(Total for Question 9 is 3 marks)

## PAST PAPER SOLUTION

Q#: 9

Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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## WORKED SOLUTION

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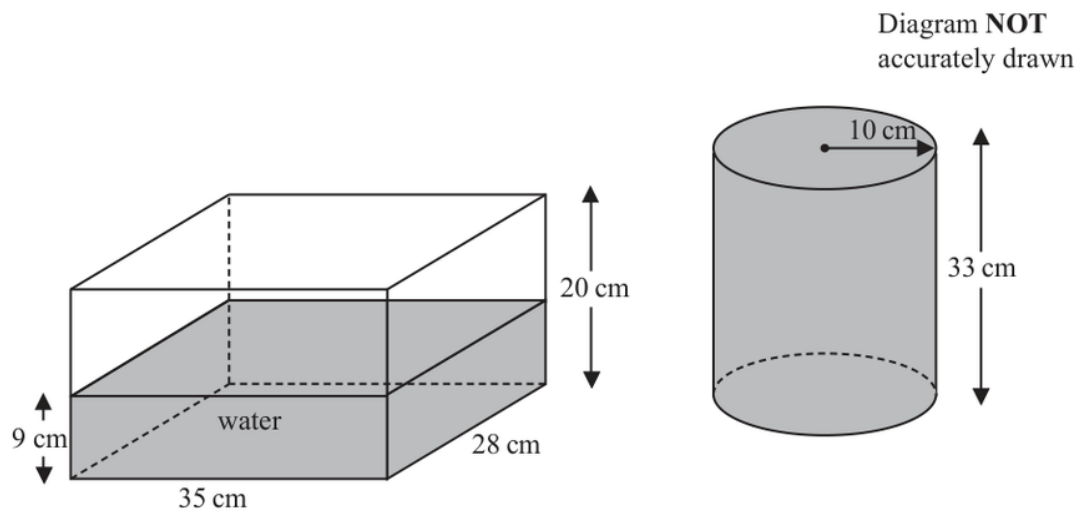
No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 10****Marks: 3****TOPIC: Volume & Surface Area**

Jun 2025 P1HR - Jun 2025 | P1HR

- 10** The diagram shows two water containers.  
One is a cuboid and one is a cylinder.



The cuboid measures 35 cm by 28 cm by 20 cm

The surface of the water in the cuboid is 9 cm above the base of the cuboid.

The cylinder has a radius of 10 cm and a height of 33 cm

The cylinder is completely full of water.

Izzy is going to pour all the water from the cylinder into the cuboid.

Show that the cuboid will not be completely full of water.

(Total for Question 10 is 3 marks)

## PAST PAPER SOLUTION

Q#: 10

Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 11****Marks: 5****TOPIC: Compound Interest & Depreciation**

Jun 2025 P1HR - Jun 2025 | P1HR

- 11** Zhou invests some money for 2 years.  
He invests \$2500 with Bank A and \$3000 with Bank B.

| Bank A                      |   |                                                    |
|-----------------------------|---|----------------------------------------------------|
| Invests \$2500              |   |                                                    |
| amount of money<br>invested | : | total amount of interest<br>after 2 years = 20 : 3 |

| Bank B                                    |  |  |
|-------------------------------------------|--|--|
| Invests \$3000                            |  |  |
| 4% per year compound interest for 2 years |  |  |

Zhou receives more **interest** from Bank A than from Bank B.

How much more?

\$ .....

(Total for Question 11 is 5 marks)

## PAST PAPER SOLUTION

Q#: 11

Marks: 5

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION

Q#: 12    Marks: 6

TOPIC: Cumulative Frequency Diagrams

Jun 2025 P1HR - Jun 2025 | P1HR

12 The table gives information about the ages of 80 people in a cinema.

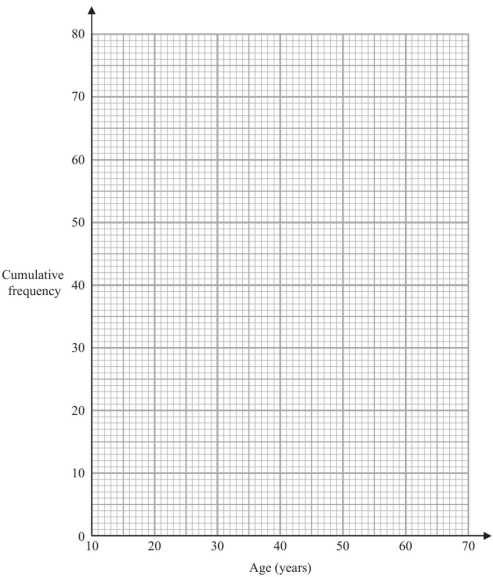
| Age ( $n$ years) | Frequency |
|------------------|-----------|
| $10 < n \leq 20$ | 12        |
| $20 < n \leq 30$ | 15        |
| $30 < n \leq 40$ | 20        |
| $40 < n \leq 50$ | 18        |
| $50 < n \leq 60$ | 9         |
| $60 < n \leq 70$ | 6         |

(a) Complete the cumulative frequency table.

| Age ( $n$ years) | Cumulative frequency |
|------------------|----------------------|
| $10 < n \leq 20$ |                      |
| $20 < n \leq 30$ |                      |
| $30 < n \leq 40$ |                      |
| $40 < n \leq 50$ |                      |
| $50 < n \leq 60$ |                      |
| $60 < n \leq 70$ |                      |

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the percentage of the 80 people who are more than 46 years of age.  
Give your answer correct to the nearest whole number.

\_\_\_\_\_ %

(3)

(Total for Question 12 is 6 marks)

## PAST PAPER SOLUTION

Q#: 12

Marks: 6

TOPIC: **Cumulative Frequency Diagrams**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 13****Marks: 2****TOPIC: Statistics Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

**13** Here are the numbers of eggs laid by some hens on each of 11 days in June.

9    10    12    15    17    18    19    19    20    25    26

Work out the interquartile range of the numbers of eggs.

**(Total for Question 13 is 2 marks)**

EA

## PAST PAPER SOLUTION

Q#: 13

Marks: 2

TOPIC: **Statistics Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 14****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jun 2025 P1HR - Jun 2025 | P1HR

**14** Work out  $6.7 \times 10^{135} + 3 \times 10^{134}$ 

Give your answer in standard form.

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**(Total for Question 14 is 2 marks)**

EA

## PAST PAPER SOLUTION

Q#: 14

Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 15****Marks: 6****TOPIC: Solving Linear Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

- 15** (a) Solve  $\frac{5a+8}{3} - \frac{2a+5}{4} = 23$   
Show clear algebraic working.

$$a = \dots\dots\dots (4)$$

- (b) Express  $\left(\frac{\sqrt{y}}{3}\right)^{-1}$  in the form  $cy^n$  where  $c$  and  $n$  are numbers to be found.

$$\dots\dots\dots (2)$$

**(Total for Question 15 is 6 marks)**

EA

## PAST PAPER SOLUTION

Q#: 15

Marks: 6

TOPIC: **Solving Linear Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 16****Marks: 2**TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

**16** Use algebra to show that the recurring decimal  $0.6\dot{1}\dot{2} = \frac{101}{165}$

(Total for Question 16 is 2 marks)

EA

## PAST PAPER SOLUTION

Q#: 16

Marks: 2

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

- 17** Prove that, for any three numbers which are consecutive multiples of 4, the difference between the square of the largest number and the square of the smallest number is always a multiple of 64

Show clear algebraic working.

(Total for Question 17 is 3 marks)

EA

## PAST PAPER SOLUTION

Q#: 17

Marks: 3

TOPIC: **Algebraic Proof**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 18****Marks: 5****TOPIC: Direct & Inverse Proportion**

Jun 2025 P1HR - Jun 2025 | P1HR

**18**  $F$  is inversely proportional to the cube of  $r$ 

$$F = 6 \text{ when } r = 2$$

(a) Find a formula for  $F$  in terms of  $r$ .....  
(3)(b) Find the value of  $r$  when  $F = 3072$ 

$$r = \text{.....}$$
  
(2)

---

**(Total for Question 18 is 5 marks)**

---

EA

## PAST PAPER SOLUTION

Q#: 18

Marks: 5

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 19****Marks: 3****TOPIC: Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

- 19** Kannika has 9 counters.  
There is a number on each counter.



Kannika puts the 9 counters in a bag.  
She takes at random a counter from the bag and does not replace the counter.  
She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters  
is less than 5

---

(Total for Question 19 is 3 marks)

EA

## PAST PAPER SOLUTION

Q#: 19

Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

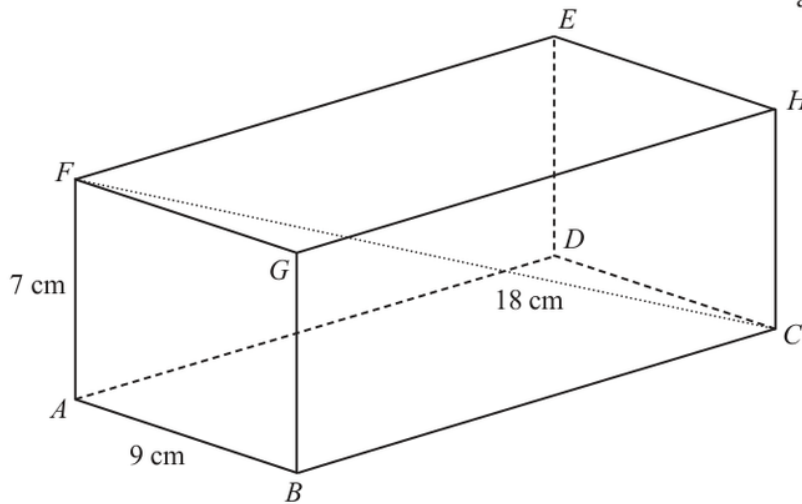
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 20****Marks: 3****TOPIC: 3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

**20** The diagram shows cuboid  $ABCDEFGH$ Diagram **NOT**  
accurately drawn

$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

Calculate the length of  $BC$ 

Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 3 marks)**

## PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

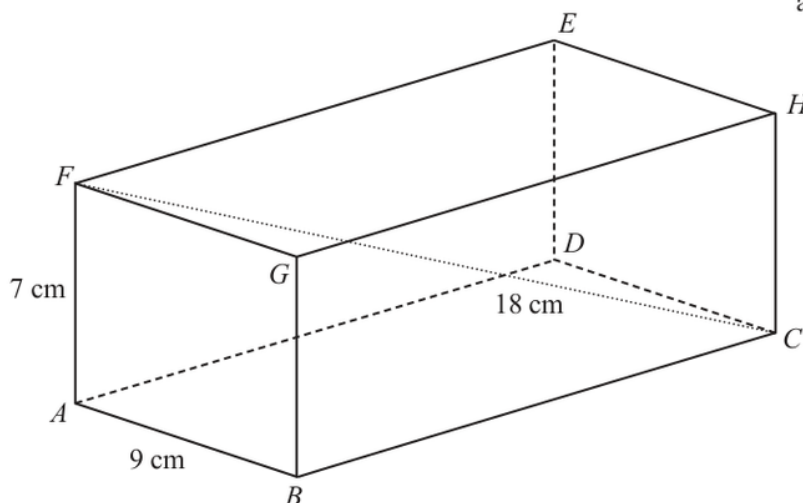
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 20** **Marks: 3****TOPIC: 3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

**20** The diagram shows cuboid  $ABCDEFGH$ Diagram **NOT**  
accurately drawn

$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

Calculate the length of  $BC$ 

Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 3 marks)**

## PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 21****Marks: 3****TOPIC: Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

- 21** The curve with equation  $y = f(x)$  has one turning point.  
The coordinates of this turning point are  $(-6, 9)$

- (a) Write down the coordinates of the turning point on the curve  
with equation  $y = f(3x)$

(....., .....)  
(1)

The curve **C** with equation  $y = g(x)$  is transformed to give the curve **S** with  
equation  $y = g(x + a) + b$

The point  $(4, -5)$  on **C** is mapped to the point  $(1, -16)$  on **S**

- (b) Write down the value of  $a$  and the value of  $b$

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

(Total for Question 21 is 3 marks)

## PAST PAPER SOLUTION

Q#: 21

Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 21****Marks: 3****TOPIC: Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

- 21** The curve with equation  $y = f(x)$  has one turning point.  
The coordinates of this turning point are  $(-6, 9)$

- (a) Write down the coordinates of the turning point on the curve  
with equation  $y = f(3x)$

(....., .....)  
(1)

The curve **C** with equation  $y = g(x)$  is transformed to give the curve **S** with  
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The point  $(4, -5)$  on **C** is mapped to the point  $(1, -16)$  on **S**

- (b) Write down the value of  $a$  and the value of  $b$

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

(Total for Question 21 is 3 marks)

## PAST PAPER SOLUTION

Q#: 21

Marks: 3

TOPIC: **Transformations of Graphs**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

**22** Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

---

**(Total for Question 22 is 5 marks)**

---

EA

## PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

**22** Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

---

**(Total for Question 22 is 5 marks)**

---

EA

## PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

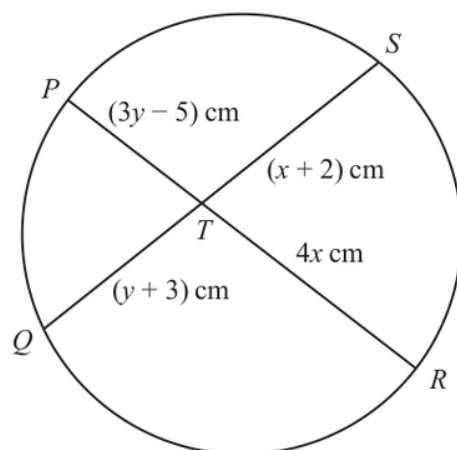
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 23****Marks: 5****TOPIC: Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

**23**Diagram **NOT**  
accurately drawn $PTR$  and  $QTS$  are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

Find an expression for  $y$  in terms of  $x$ 

$$y = \dots\dots\dots$$

**Total for Question 23 is 5 marks)**

## PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

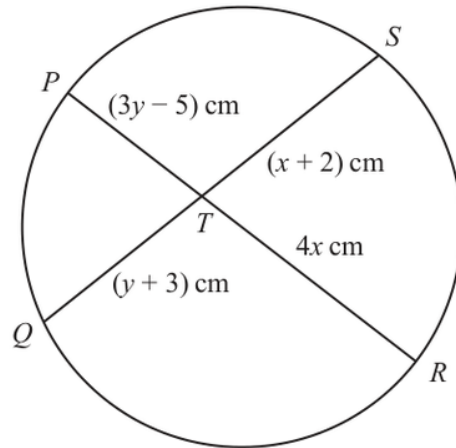
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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 23****Marks: 5****TOPIC: Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

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accurately drawn $PTR$  and  $QTS$  are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

Find an expression for  $y$  in terms of  $x$ 

$$y = \dots\dots\dots$$

**Total for Question 23 is 5 marks)**

## PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Circle Theorems**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 24****Marks: 6****TOPIC: Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

**24** The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where  $x$  and  $y$  are constants.

The sum of the first 9 terms of the series is 216

Find the value of  $x$  and the value of  $y$ 

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

**(Total for Question 24 is 6 marks)**

## PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

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No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 24****Marks: 6****TOPIC: Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

**24** The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where  $x$  and  $y$  are constants.

The sum of the first 9 terms of the series is 216

Find the value of  $x$  and the value of  $y$   
Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

**(Total for Question 24 is 6 marks)**

## PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Sequences**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

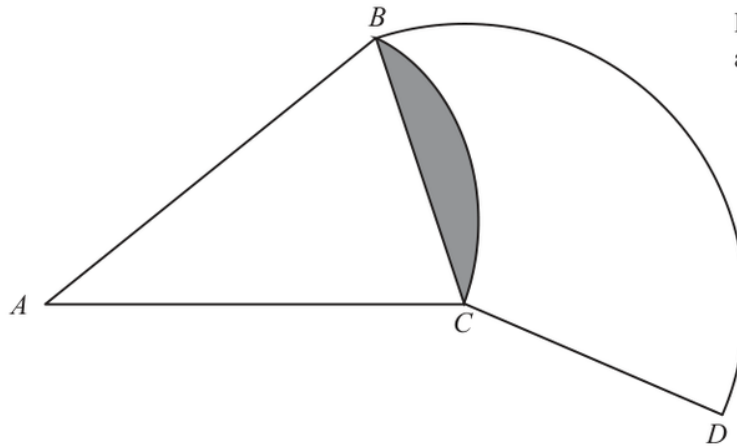
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

**25**Diagram **NOT**  
accurately drawn $BAC$  is a sector of a circle, centre  $A$  $BCD$  is a sector of a circle, centre  $C$ Angle  $BAC = 40^\circ$ Angle  $BCD = 130^\circ$ Area of shaded segment =  $28 \text{ cm}^2$ Find the length of the arc  $BD$ 

Give your answer correct to 3 significant figures.

---

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..... cm

**(Total for Question 25 is 6 marks)**

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## PAST PAPER SOLUTION

Q#: 25

Marks: 6

TOPIC: **Circles, Arcs & Sectors**

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## WORKED SOLUTION

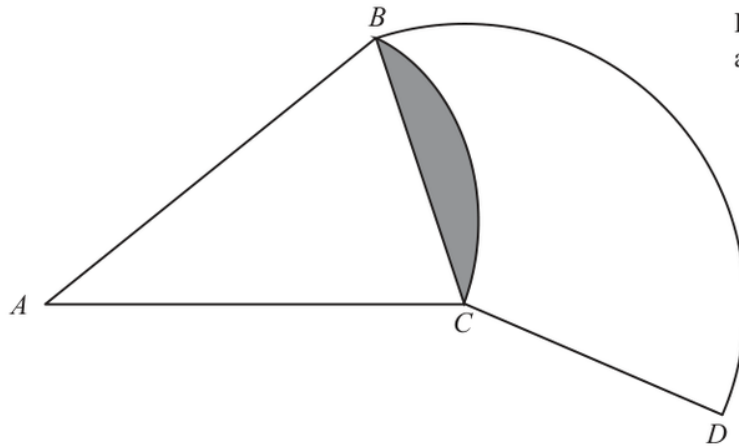
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 25****Marks: 6****TOPIC: Circles, Arcs & Sectors**

Jun 2025 P1HR - Jun 2025 | P1HR

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---

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..... cm

**(Total for Question 25 is 6 marks)**

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## PAST PAPER SOLUTION

Q#: 25

Marks: 6

TOPIC: **Circles, Arcs & Sectors**

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## WORKED SOLUTION

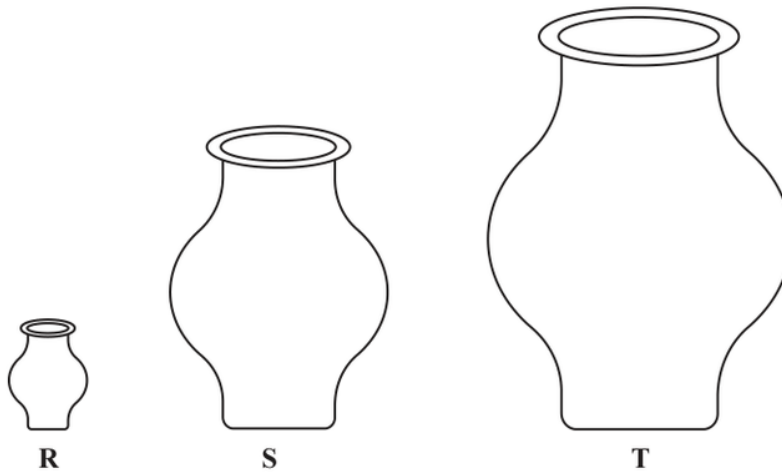
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 26****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

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**26** **R**, **S** and **T** are three similar vases.Diagram **NOT**  
accurately drawnThe volume of vase **S** is 72.8% more than the volume of vase **R**The height of vase **R** is  $h$  cmThe height of vase **T** is  $6h$  cmThe surface area of vase **S** is  $A$  cm<sup>2</sup>The surface area of vase **T** is  $kA$  cm<sup>2</sup>Work out the value of  $k$  $k = \dots\dots\dots$ **(Total for Question 26 is 4 marks)**

## PAST PAPER SOLUTION

Q#: 26

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

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## WORKED SOLUTION

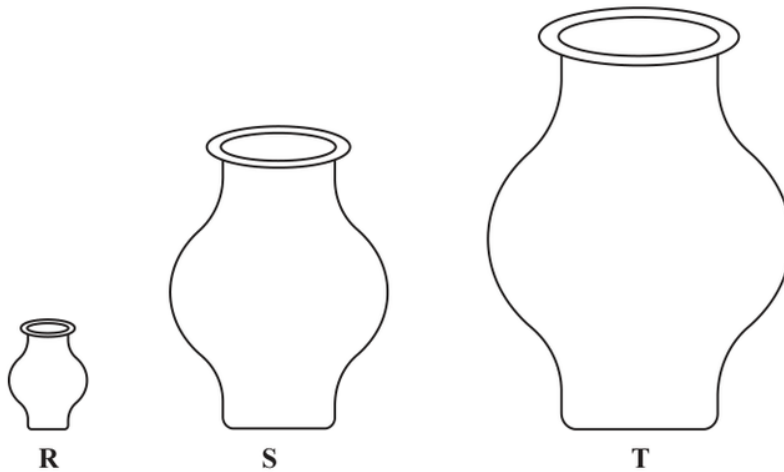
*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA

**PAST PAPER QUESTION****Q#: 26****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

**26** **R**, **S** and **T** are three similar vases.Diagram **NOT**  
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## PAST PAPER SOLUTION

Q#: 26

Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jun 2025 P1HR - Jun 2025 | P1HR

## WORKED SOLUTION

*Dr. Eslam Ahmed*

No worked solution is saved yet.

EA